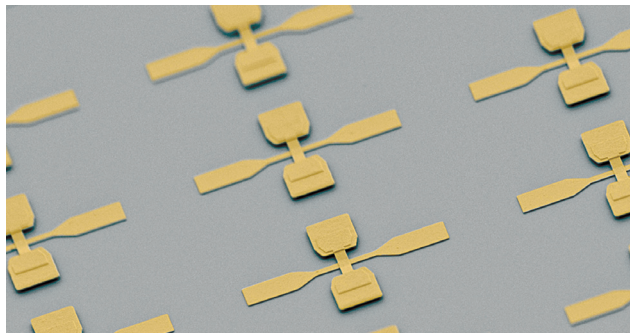


New fabrication technique uses sub-nanometre molecular layers

MIT researchers have developed a scalable fabrication platform that integrates fragile molecular materials into functional electronic devices without damaging them. The



A false-colour electron microscope image shows the devices. Using the new scalable fabrication technique, researchers fabricated more than 1000 devices using sub-nanometre molecular layers (Image: Courtesy of the researchers)

two-step process uses conventional semiconductor manufacturing to build device components before introducing molecular layers, followed by nanoscale forces that create self-aligned electrical contacts. Demonstrated in Cambridge, Massachusetts, the technique produced more than 1000 devices using sub-nanometre molecular layers and enabled interconnected molecular memory arrays. The approach could support next-generation computing, sensing, photonics, and quantum technologies by combining the scalability of chip manufacturing with the precision needed to integrate atomic-scale materials.

AI model brings agentic AI to PCs

Meta Superintelligence Labs has released Muse Glimmer, a 30-billion-parameter open-weights AI model designed to bring agentic AI workloads to PCs. The model supports long-context processing, persistent memory, tool use, decision-making, and multi-step workflows, and can run locally on AMD Ryzen AI Max+ systems and Radeon AI PRO R9700 GPUs. Developed by a team at Meta, the model targets applications requiring autonomous task execution while keeping data on-device. Local processing can reduce cloud dependence, improve data control, and eliminate recurring token costs. The Apache 2.0 licence also enables commercial modification and redistribution.

Portable device detects water contaminants in minutes

Researchers at IIT Guwahati have developed E-Eye, a portable optical device that detects water contaminants and delivers digital results for on-site monitoring. The system analyses colour changes produced by chemical reactions,



IIT Guwahati develops 'E-Eye' for rapid detection of cancer-causing heavy metals and other toxicants (Credit: www.iitg.ac.in)

then uses optical sensing to quantify pollutants including arsenic, lead, iron, chromium, fluoride, and E. coli. Developed by a team in Guwahati, the device matched conventional laboratory instruments during evaluations while adding wireless data transmission. Its prototype costs about ₹3500, with tests costing around ₹2. E-Eye could support drinking-water checks, wastewater monitoring, food

testing, and environmental analysis, with solar-powered, cloud-connected versions under development.

Autonomous system for monitoring and cleaning solar plants

Researchers at the National Institute of Technology (NIT) Rourkela have developed an AI-powered autonomous system that monitors solar panels and performs cleaning only when required. The patented platform combines federated learning,



NIT Rourkela develops autonomous system and method for monitoring and cleaning solar plants

edge computing, fault detection, autonomous cleaning, and predictive maintenance. It shares encrypted model updates instead of raw data, improving privacy, cybersecurity, bandwidth

efficiency, and scalability. Developed in Rourkela, the system can serve utility-scale, floating, rooftop, industrial, and remote solar installations. By enabling selective, condition-based cleaning, it can reduce water use, labour, maintenance costs, and power losses caused by debris.

SRAM chip slashes space power

A memory-centric SRAM chip could cut onboard AI power for future space telescopes by up to 59 times compared with GPU-based systems. Developed by engineers at the University of Michigan in Ann Arbor, the architecture performs computations near stored data, reducing energy-hungry data transfers. Simulations indicate power demand could fall from 3000W to 51W, potentially reducing spacecraft mass from 1100kg to 193kg and saving \$430 million over 25 years. The design targets real-time wavefront correction, adaptive optics, and exoplanet imaging, while offering potential for satellites, deep-space probes, and autonomous spacecraft.